



All for One and One for All – Collaboration in Computing Education: Policy, Practice, and Professional Dispositions

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ABSTRACT

The ITiCSE '23 final keynote raised teaching soft skills, or professional dispositions, to help students face challenges in modern programming. This project addresses helping computing students develop professional dispositions through collaborative learning (CL) since some in the industry observe entry-level engineers struggling due to their fragile professional dispositions. We are motivated to understand professional expectations from entry-level engineers and present the academia-industry gap to support practitioners and researchers in advancing CL in Computing Education, encouraging positive curricula and policy changes that promote DEIA. We will present CL practices alongside their supported professional dispositions to assist practitioners in adoption. We will present the academia-industry gap in CL for future research opportunities, helping researchers advance CL practices to integrate professional dispositions the industry expects from entry-level engineers.

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CCS CONCEPTS

• **Software and its engineering**; • **Social and professional topics** → **Gender**; • **Human-centered computing** → *Collaborative and social computing*;

KEYWORDS

Collaborative Learning, Professional Dispositions, Computing Education

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1 INTRODUCTION

Software development is typically a collaborative process to create applications and services, and the importance of skill in collaboration has received recent attention; for example, the ITiCSE '23 final keynote raised soft skills needed by students to face challenges in modern programming. Soft skills are congruent to *professional dispositions*, behavioural patterns that demonstrate professional values through peer interactions [2].

Unfortunately, individuals in entry-level roles struggle with communication and collaboration due to fragile technical and professional dispositions skills [9]. Apprehension in practising professional dispositions may be why individuals in entry-level roles struggle to apply them in the workplace.

2 MOTIVATION

We are motivated to support practitioners in identifying and adopting collaborative learning (CL) practices that encourage Diversity, Equity, Inclusion, and Accessibility (DEIA) and support positive, inclusive CL for students to practise and develop professional dispositions. We want to understand the struggles individuals in entry-level roles have with professional dispositions and raise awareness within the Computing Education community on the types of professional dispositions the industry expects so that academia can improve the support of these dispositions through CL. We aim to identify the academia-industry gap, giving the Computing Education community opportunities to adapt computing curricula and policies to support professional dispositions that align with the industry's expectations, guided by the following questions:

- How do computing educators incorporate the teaching and learning of professional dispositions into tertiary institutions?
- How can educators utilise these teaching and learning practices to foster DEIA mindsets?
- How do collaborative learning activities promote professional dispositions and cultivate DEIA mindsets?
- What non-technical skills and knowledge are anticipated by the industry for individuals in entry-level roles?
- How do the professional dispositions applied in academic learning environments align with the industry's expectations and computing education policy?

3 BACKGROUND

With increased interest in students' professional dispositions, computing curricula, like CC2020 [2], recommend using *collaborative learning* (CL), an educational theory that brings students together to master learning concepts. In industry, professional dispositions help teams create software effectively, which makes these skills highly desirable to employers [7]. However, professional dispositions are not commonly listed in course syllabi as learning outcomes demonstrating an academia-industry gap in expectations of these skills from students.

Research focusing on DEIA efforts during CL found mixed results. Some research on peer instruction (PI) shows improvements in learning gains and sense of belonging, especially for underrepresented minority students (URMs) [4]. CL techniques like PI, process oriented guided inquiry learning (POGIL), and pair programming can help create a positive student community for female students to find support and peer study groups [6]. However, the impact and experiences on individuals in diverse student teams are more complex than team collaboration in the industry. Sometimes, the learning experience can feel non-inclusive and inequitable. URMs perceive they are not included in decision-making processes, generating frustration [5]. Within mixed-gender student teams, males sometimes do not let females fully participate. Between URMs and those students in the majority, there are notable disparities in their

perceptions of CL. For example, a CS1 CL study [1] showed URM students perceived lower instructor support and formed lower opinions about the benefits of CL. A week-long STEM camp showed the unfavourable effect of increasing the confidence of males but decreasing that of females [10].

When examining student collaboration during CL, there are inequitable efforts within mixed-gender teams. A study [3] analysed the communication of female and male students, showing significant statistical results with female students' initiated communication. Female students engaged more leadership, project coordination, and monitoring skills to complete the project. Disparities in group responsibilities may contribute to URMs, like women, having waning interests in computing after their first year [8].

4 METHOD

This study has a three-pronged approach: (1) a literature review identifying collaborative learning (CL) in Computing Education, analysing CL practices using clustering analysis that promotes DEIA; (2) a literature review examining the industry's expectations of entry-level roles' professional dispositions, categorising these dispositions to identify the most desired by organisations; and (3) a survey and interviews giving a multi-national perspective on professional dispositions software organisations seek from entry-level roles. The methodology enables us to report the academia-industry gap in professional dispositions and how including these dispositions in recommended computing curricula, like CC2020, aligns with our findings.

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